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GaitGuard: TinyML-based fall risk assessment for older adults using foot-mounted IMU



Yilu Huang · XIAO ESP32-S3 + LSM6DS3 · Edge Impulse · GSTRIDE v3 dataset

1 Problem

Falls are the leading cause of injury-related death in adults over 65. Yet most devices only respond after a fall happens - reactive.

> GaitGuard is proactive: by continuously analyzing gait patterns on-device, it detects deterioration before an incident occurs and alerts the user immediately.

Primary users
Adults 65+

Secondary users
Family members
Strangers

2 Hardware

- XIAO ESP32-S3
- ERM motor
  P = Medium
- 1000mAh LiPo
- Buzzer
  P = Medium
- LSM6DS3 IMU
  104Hz, 6-axis, I2C
- ±16g / ±2000dps
  Matches GSTRIDE config



3 ML Pipeline

Dataset & Pre-processing

- GSTRIDE v3 · 148 elderly subjects · 70–98 yrs · foot-mounted IMU
- 5s window · no overlap · subject-level 80/20 split (no data leakage)
- 6 axes: accX/Y/Z + gyrX/Y/Z @ 104Hz → 3102 features per window
- Binary labels: faller / nonfaller · class_weight {faller: 1.5}

Model evolution

Model	DSP block	Test accuracy	Faller recall	AUC	Notes
Dense 20→10	FFT (2s)	57.4%	0.63	0.66	Baseline
Dense 20→10	Wavelet db4 (5s)	63.5%	0.58	0.65	Larger window
Multi-scale CNN	Raw data (5s)	**74.5%**	**0.821**	**0.74**	Final model

Final model architecture — multi-scale CNN

Input (3102, 1) → Conv1D 8 filters, k=4, BN + ReLU, MaxPool /4, Drop 0.25 → Conv1D 16 filters, k=8, BN + ReLU, MaxPool /4, Drop 0.25 → GAP Global avg pooling → Dense 16 units, ReLU, Drop 0.4 → Output 2 classes softmax INT8

84ms inference · 51.4KB RAM · EON compiler · Quantized INT8

Confusion matrix

	FALLER	NONFALLER	UNCERTAIN
FALLER	82.1%	17.9%	0%
NONFALLER	33.9%	66.1%	0%
F1 SCORE	0.77	0.71	

4 On-device inference pipeline

IMU sampling
104Hz · 6 axes · 5s window = 3102 floats

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Static detection
acc_ms < 100 → skip inference (silent)

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CNN inference
84ms · 51.4KB RAM · INT8

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Two-layer smoothing
Short-term 5 windows (25s) → risk score 6 windows (30s)

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LOW	MEDIUM	HIGH
< 0.55	0.55–0.75	⩾ 0.75
Silent	Vibration	Buzzer ×3

5 Limitations & future work

- Thresholds not calibrated with real elderly users
- Val–test gap (91%→74%) — generalization difficulty

1. CNN beats feature engineering: Raw data + CNN: 63.5% → 74.5% accuracy, 0.58 → 0.821 faller recall
2. TinyML on-device is viable: 84ms inference, 51.4KB RAM, no cloud — real-time on ESP32-S3

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- BLE caregiver notification + mobile app
- Flexible material + smart insole form for improved comfort